

A Reactive Swarm Controller for the ARGoS Tunnelling Task

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Abstract. This report presents a reactive controller for the ARGoS tunnelling project. The implementation combines local repulsion and attraction vectors, a phase-based state machine, and a small set of arena cues to coordinate a swarm of foot-bots without centralized control. The controller uses range-and-bearing communication to maintain spacing, proximity and light sensing to bias motion, and motor-ground readings to detect the black target zone. The current version implements three main phases: initial orientation, grouping, and tunnel progression, followed by a black-zone behavior that keeps the robots inside the target region once they enter it. The report also documents the project requirements, the design choices made in the controller, and the evaluation protocol to be used for scaling experiments. Experimental figures and summary tables are left as explicit placeholders where results still need to be added.

Keywords: swarm robotics · ARGoS · Lua · tunnelling · foot-bot

1 Introduction

The project asks for a cooperative collective behavior able to solve the tunnelling task in ARGoS: the swarm must move through an obstacle-filled arena, reach the target zone on the right, and avoid bringing obstacles into the target. The score is defined as the number of robots in the target minus the number of obstacles in the target at the end of the run. The assignment also requires repeated runs with different random seeds, performance summaries in the form of boxplots, and a scalability analysis across swarm sizes.

The implementation discussed in this report is written in Lua for the foot-bot platform and follows the local sensing and actuation model described in the course notes for Lua and the foot-bot API [1, 2]. The design is intentionally reactive: each robot reacts to nearby peers, light, floor color, and obstacles rather than relying on a global map or centralized coordination.

2 Problem Definition and Requirements

The arena is a rectangular environment with a white starting area on the left, an obstacle field in the middle, and a black target area on the right. The project brief specifies a 1,000 second run, corresponding to 10,000 ARGoS steps, although a shorter runtime may be used if justified in the report.

The available sensors and actuators include proximity, light, motor-ground, range-and-bearing, colored-blob omnidirectional camera, wheels, LEDs, gripper, and turret. The brief emphasizes cooperative behavior, use of swarm robotics principles, repeated runs over 10 seeds, and a scaling study over several swarm sizes. In addition to the final score, the project data file records the number of robots and obstacles in the target at each step.

2.1 Evaluation Criteria

The main quantities to report are:

- final score as a function of swarm size,
- number of robots in the target over time,
- number of obstacles in the target over time,
- variability across the 10 repetitions for each configuration.

3 Controller Overview

The controller is organized as a state machine with four phases: orientation, grouping, tunnel progression, and black-zone behavior. The transition logic is driven by local conditions such as the number of neighbors at a target spacing, floor color, and whether the robot has already entered the black target zone.

At a high level, the controller combines three vector sources:

- a Lennard–Jones interaction over range-and-bearing messages,
- a light-seeking vector from the ambient light sensor,
- obstacle repulsion and tangential corrections from proximity readings.

The implementation also uses robot LEDs to expose the current phase, which is useful during debugging and qualitative analysis. Per-robot logs are written to the `logs/` directory.

3.1 Orientation Phase

In the initial phase, robots bias motion toward the light while maintaining some distance from nearby robots and obstacles. The controller uses the light vector, the Lennard–Jones interaction, and a small obstacle tangential correction to avoid overcrowding at the start. After a fixed number of steps, the robot moves to the grouping phase.

3.2 Grouping Phase

The grouping phase uses the range-and-bearing system to enforce a loose swarm structure. The code computes a Lennard–Jones potential at the current peer distance and counts neighbors in a target interval around 80 cm. Once enough neighbors are found consistently for a sufficient number of steps, the robot switches to the tunnel phase.

The controller also computes a leader vector from robots that already advanced into later phases. This creates a simple local bootstrap effect: robots that are already closer to the target influence nearby robots to follow them.

3.3 Tunnel Phase

The tunnel phase reduces the target spacing to 60 cm and strengthens the attraction toward robots already in the tunnel. The phase uses a combination of Lennard–Jones interaction, light attraction, and leader following. If an obstacle is directly ahead, the controller adds a sideways correction so the robot can move around the obstacle field instead of pushing straight through it.

This phase is the main navigation behavior of the current implementation. It is designed to help the swarm advance collectively through the clutter while still preserving a coherent formation.

3.4 Black-Zone Phase

When enough floor sensors detect black for several consecutive steps, the robot enters the black-zone state. In this state, the controller keeps the robot in the target region and uses a randomized heading segment to avoid getting stuck on the boundary. If the robot sees the green beacon of another robot already in the black zone, it will bias motion toward that robot as a simple local cue.

The motor-ground readings are central here: they determine when the robot has reached the target and when it should leave the boundary-following motion.

4 Implementation Details

4.1 Sensors and Local Rules

The controller uses the following signals:

- **Proximity sensors:** obstacle repulsion, obstacle tangential motion, and wall/robot blocking.
- **Light sensors:** orientation toward the ambient cue.
- **Motor-ground sensors:** detection of the black target zone.
- **Range-and-bearing:** local communication and Lennard–Jones spacing.
- **Colored-blob camera:** detection of a green beacon emitted by robots already in the black zone.
- **LEDs:** visual phase feedback during simulation.

The wheel command is computed from the target vector angle with a simple proportional steering law. The result is clamped to the allowed wheel speed. This keeps the behavior easy to reason about and consistent with the reactive style encouraged in the project brief.

4.2 State Transitions

The state machine is driven by local counts and thresholds rather than by global knowledge. The most relevant conditions are:

- a minimum number of neighbors in the target spacing interval,
- a minimum duration during which that grouping condition must hold,
- a consecutive black-floor counter before entering the black zone,
- a beacon-based check to bias motion once inside the target region.

4.3 Current Limitation

The controller contains an obstacle-grabbing sequence using the gripper and turret, but that path is currently disabled in the implementation because the corresponding handler returns immediately. As a result, the current version focuses on navigation, grouping, and black-zone stabilization rather than on physical obstacle transport. This should be stated explicitly in the final report, since the project brief expects the gripper and turret to be used for a full tunnelling solution.

5 Experimental Protocol

The project brief requests 10 runs per configuration with different random seeds. The report should present boxplots of the final score for each swarm size, along with time-series plots for the number of robots in the target and the number of obstacles in the target.

5.1 Planned Settings

Table 1. Planned experimental settings. Replace placeholders with the final tested configurations.

Configuration	Value	Notes
Run duration	10,000 steps	Default setting
Seeds per setting	10	Required
Swarm sizes	<insert sizes>	Prefer higher resolution
Baseline	<insert baseline if any>	Optional comparison

5.2 Data to Report

Table 2. Summary placeholders for experiment outputs. Replace with computed statistics once the runs are available.

Swarm size	Mean score	Std. dev.	Median score
<n1>	<value>	<value>	<value>
<n2>	<value>	<value>	<value>
<n3>	<value>	<value>	<value>

6 Results

The following figures are placeholders for the final analysis. They should be replaced by the actual plots generated from the 10 repeated runs per swarm size.

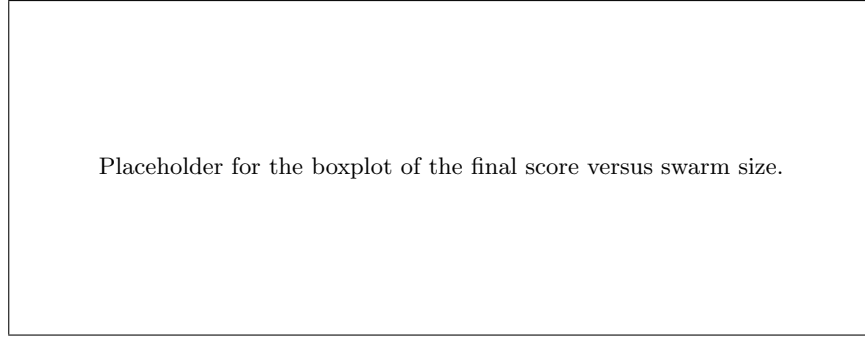


Fig. 1. Final score as a function of swarm size. Each box should summarize 10 repetitions.

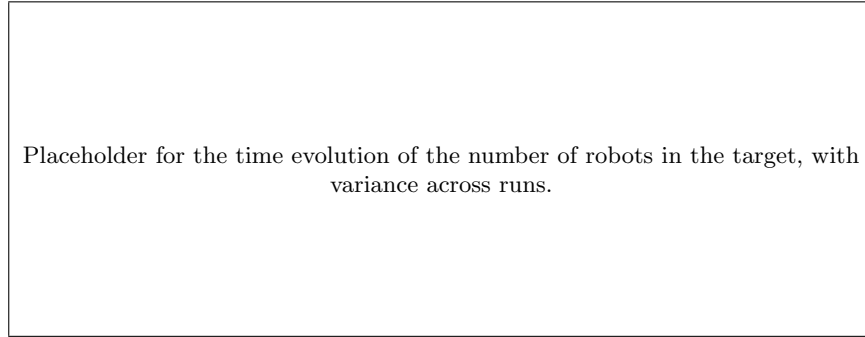


Fig. 2. Robots in the target over time for representative swarm sizes.

6.1 Discussion Points for the Final Draft

Once the data are inserted, the discussion should answer the following:

- How does performance scale with the number of robots?
- Is the performance monotonic, or do some sizes perform worse than nearby sizes?
- Do some swarm sizes exhibit higher variance than others?

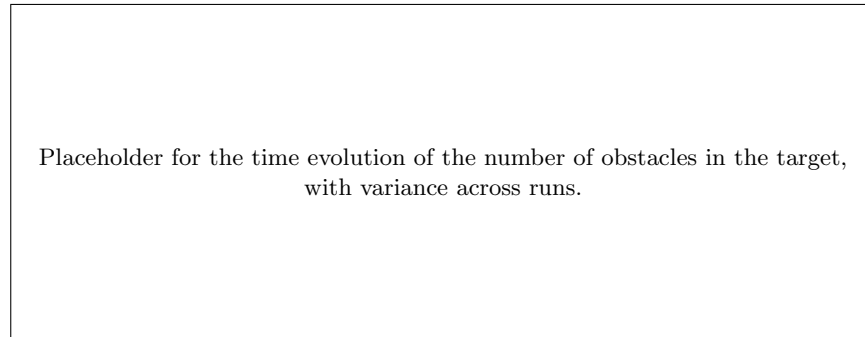


Fig. 3. Obstacles in the target over time for representative swarm sizes.

- Are the observed trends consistent with the controller’s local rules and phase transitions?

7 Conclusion

The current controller implements a clear reactive structure that is aligned with the swarm robotics principles emphasized in the course: local sensing, distributed coordination, and simple rules that produce collective motion. The report draft is complete enough to receive the final experimental plots and statistics, but the obstacle-transport capability still needs to be finished or explicitly presented as an open limitation if no further code changes are made.

References

1. H414 – Swarm Intelligence: Programming the Robots with Lua. Course reference document.
2. H414 – Swarm Intelligence: Sensor and actuator reference for the foot-bot. Course reference document.
3. H414 – Swarm Intelligence: Evaluation and Final Project of INFO-H-414 Swarm Intelligence, Second Session.
4. Springer: Lecture Notes in Computer Science style and sample paper documentation.